



Here's the latest CodeChef.com Challenge, your monthly dose of coding puzzles from India's biggest coding contest!

Solve this month's puzzle, and you could be one of the three lucky people to win a cash prize of Rs 1,000 each!

The June edition's puzzle:

Our chef planned to open a new restaurant. He went to the market and bought some plates and bowls. He hired a person to take the plates and bowls to the restaurant. The man broke some of them on the way to the restaurant, and exactly two-thirds of the bowls and three-fourths of the plates got delivered safely. If an equal number of bowls and plates were found to be okay, what fraction of the total number of items got damaged?

The Solution:

Let's suppose the chef bought x plates and y bowls.

The hired person delivered $3/4$ ths of the plates, i.e., $(3x/4)$ and $2/3$ ths of the bowls, i.e., $(2y/3)$.

Hence, the damaged plates = $x - (3x/4) = x/4$.. (1)

And the damaged bowls = $y - (2y/3) = y/3$.. (2)

An equal number of bowls and plates are found to be intact, i.e., $(3x/4) = (2y/3)$

This implies $x = (8y/9) \dots (3)$

The fraction of the total items that got damaged = $(x/4 + y/3) / (x + y)$...using Equations (1) and (2)

$$= ((2y/9) + (y/3)) / ((8y/9) + y) \text{ ..using Eq (3)} \\ = 5/17$$

Hence, $5/17$ of the total items got damaged.

And the winners this month are:

- Durgadoss Ramanathan
- Sudhanshu Gupta
- Satyabrata Satpathy

Here's your CodeChef Puzzle of the Month:

There is a village that is situated on a hill. It doesn't have any resources of its own and hence everything has to be brought up from the city market. Every month the government sends a cart down to the market to buy what's required by the people. While returning to the village, the cart gets overloaded every time. The cart is pulled by a single horse at a time. One horse can pull the cart for 10 km and can be used once a day only. Three horses are used initially, and after they pull the cart, the cart has to stop so that the animals can recoup for the next day's journey. But due to the load and an inefficient braking system, the cart rolls back 10 km during the journey uphill. During the last day of the journey, one more horse joins the team to pull the cart. How many days will it take the cart to reach the village if the distance between the market and the village is 400 km?

Looking for a little extra cash?

Solve this month's CodeChef Challenge and send in your answer to codechef@efyindia.com or lfy@codechef.com by August 15, 2012. Three lucky winners get to win some awesome prizes both in cash and merchandise! You can also participate in the contest by visiting the online space for the article at <https://www.facebook.com/LinuxForYou>

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About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. Log on to CodeChef.com for the August Challenge that takes place from the 1st to the 11th, to win cash prizes up to Rs 20,000. So keep visiting CodeChef and participate in the multiple programming contests that take place on the website throughout the month!